

[Courseware \(/courses/HumanitiesScience/StatLearning/Winter2014/courseware\)](/courses/HumanitiesScience/StatLearning/Winter2014/courseware)[Course Info \(/courses/HumanitiesScience/StatLearning/Winter2014/info\)](/courses/HumanitiesScience/StatLearning/Winter2014/info)[Discussion \(/courses/HumanitiesScience/StatLearning/Winter2014/discussion/forum\)](/courses/HumanitiesScience/StatLearning/Winter2014/discussion/forum)[Wiki \(/courses/HumanitiesScience/StatLearning/Winter2014/course_wiki\)](/courses/HumanitiesScience/StatLearning/Winter2014/course_wiki)[Progress \(/courses/HumanitiesScience/StatLearning/Winter2014/progress\)](/courses/HumanitiesScience/StatLearning/Winter2014/progress)[Lecture Slides \(pdf\) \(/courses/HumanitiesScience/StatLearning/Winter2014/ba1951b8f66c4cdca2fcaabdc91b792/\)](/courses/HumanitiesScience/StatLearning/Winter2014/ba1951b8f66c4cdca2fcaabdc91b792/)[R Sessions \(/courses/HumanitiesScience/StatLearning/Winter2014/0d68641c19484ef9aa6aeea03426dc68/\)](/courses/HumanitiesScience/StatLearning/Winter2014/0d68641c19484ef9aa6aeea03426dc68/)

4.6.R1 (1/1 point)

Which of the following is NOT a linear function in x:

- ☐ $f(x) = a + b^2 x$
- ☐ The discriminant function from LDA
- ☐ $\delta_k(x) = x \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$
- ☐ $\text{logit}(P(y = 1|x))$ where $P(y = 1|x)$ is as in logistic regression
- ☒ $P(y = 1|x)$ from logistic regression ✓

EXPLANATION

$P(y = 1|x)$ from logistic regression is not linear because it involves both an exponential function of x and a ratio. Notice that $f(x) = a + b^2 x$ is not a linear function of b, but is a linear function of x.

Hide Answer

You have used 1 of 5 submissions

